GREENFIELD DEVELOPERS PVT LTD
BIM Coordination — Request for Information

RFI #RFI-PRJ-2024-GT-001-CLH-003

Issued: 21 September 2026
Auto-generated by BIM Clash Engine

● HIGH SEVERITY | HARD CLASH | PLUMBING vs STRUCTURAL | LEVEL 4 | GRID F8

PROJECT INFORMATION

Project Name Greenfield Tower — Block A	**Project Number** PRJ-2024-GT-001	**Client** Greenfield Developers Pvt Ltd
Location Sector 62, Noida, Uttar Pradesh	**RFI Number** RFI-PRJ-2024-GT-001-CLH-003	**Date Issued** 21 September 2026

RFI SUBJECT

RFI-003: Uncoordinated Penetration of 110mm UPVC Sanitary Stack ST-04 through 200mm RCC Slab at Grid F8, Level 4

CLASH ELEMENT DETAILS

Field	Element A — Plumbing	Element B — Structural
Revit ID	REV-PL-002219	REV-STR-004562
Category	Pipe	Floors
Family	Sanitary Drain — 110mm UPVC	RCC Slab 200mm
Size	110mm diameter	200mm thick
System	Sanitary Drainage — Stack ST-04	—
Level	Level 4	Level 4
Grid Ref	F8	F8
Clash Point	X: 67.890 Y: 12.340 Z: 12.200	Penetration: -0.110m

TECHNICAL DESCRIPTION

A hard clash has been identified in the Level 4 structural model where the 110mm diameter UPVC sanitary drain stack (ST-04) intersects the 200mm thick RCC slab at Grid F8 without a pre-installed structural sleeve. The current model indicates a direct penetration depth of -0.110m, which compromises the structural integrity of the slab and lacks the necessary fire-stop or waterproofing provisions required for vertical drainage systems passing through structural elements.

IMPACT ASSESSMENT

Failure to resolve this clash will result in rework during structural casting or post-construction core drilling, leading to potential delays in the Level 5 structural sequence and increased risk of water leakage or structural weakening at the penetration point.

APPLICABLE CODES & STANDARDS

- IS 456:2000 (Plain and Reinforced Concrete - Code of Practice)
- NBC 2016 Part 4 (Plumbing and Drainage)
- IS 15380:2003 (Code of Practice for Design and Construction of Water Supply and Sanitary Drainage Systems)
- IS 800:2007 (General Construction in Steel - for reference on penetration limits if steel beams are involved, though primary is concrete)

PROPOSED RESOLUTION

Install a pre-cast concrete sleeve or a heavy-gauge steel sleeve with a diameter of 150mm (minimum 40mm clearance around the 110mm pipe) embedded within the 200mm RCC slab at Grid F8. The sleeve must be sealed with non-shrink grout or approved fire-stop material to maintain structural continuity and prevent water migration.

Option 1: Route the sanitary stack ST-04 through an adjacent grid line (e.g., F7 or F9) where the slab thickness or structural layout permits easier penetration, subject to hydraulic slope verification.
Option 2: Utilize a post-construction core-drilling method with a certified structural engineer's approval, followed by the installation of a structural repair sleeve and waterproofing membrane, if pre-casting is no longer feasible due to construction progress.

INFORMATION REQUIRED

Please provide the approved structural detail for the sleeve size, material, and anchorage requirements for the 110mm pipe penetration at Grid F8. Confirm if a pre-cast sleeve is mandatory or if a post-construction core drill is permissible with specific reinforcement detailing. Additionally, specify the required fire-stop or waterproofing material specifications for the annular space.

ROUTING & RESPONSE DEADLINE

Assigned To	Role	Email	Response Required
Priya Sharma	Senior Structural Engineer	devendragupta56789@gmail.com	3 days

Issued By: Devendra Gupta | BIM Automation Engineer
Company: Greenfield BIM Team | **Date:** 21 September 2026 | **Auto-generated:** 21-Sep-2026 09:25